

STATEMENT OF WORK

CRYOTECH E36 LIQUID AND SOLID RUNWAY DEICER

THE CONTRACTOR SHALL DELIVER SUPPLIES TO THE GOVERNMENT AND OR PERFORM THE SERVICES CALLED FOR HEREUNDER AS FOLLOWS

1. SPECIFICATIONS FOR LIQUID RUNWAY DEICER:

- 1.1. Fluid, Deicing/Anti-icing for Runways and Taxiways
- 1.2. Cryotech E36 brand Potassium Acetate based - in addition to requirements set by AMS 1435 A full copy of the latest revision of the AMS 1435 specification can be found at SAE www.sae.org
- 1.3. Specifications cover a potassium-acetate based deicing/anti-icing fluid in the form of a concentrate liquid meeting FAA-approved specifications SAE AMS 1435 ("The Fluid").
- 1.4. The Fluid shall be minimum 50% potassium acetate by weight with a freezing point of -76°F (-60°C) or lower (method: ASTM DI 177). The Fluid shall not contain Urea, ethylene or propylene glycol.
- 1.5. The vendor shall provide comprehensive, on-site training and/or technical briefing to personnel on proper use and application rates if required.
- 1.6. The vendor shall provide the name and location of The Fluid production site and a copy of independent laboratory certification of The Fluid to AMS 1435. The certification shall be dated within 24 months of bid submittal. If a license of subcontractor produces The Fluid, all testing shall be required on The Fluid produced at the site as if The Fluid were being initially qualified. The vendor must provide documentation that the licensee or subcontract conforms to AMS 1435
4.4.3. Whenever a fluid is to be produced by a licensee or subcontractor, all testing shall be required on fluid produced at the site, prior to initial shipment. This information is to be provided with bid submittal.

- 1.7. The vendor shall provide written technical information covering at minimum: The Fluid's environmental impact, SDS, handling storage and application guidelines. This information is to be provided with bid submittal.
- 1.8. The vendor shall provide a complete acute aquatic toxicity test report for The Fluid. The report shall be by an independent party and state compliance with U.S. EPA Test Methods 40 Code of Federal Regulations Parts 797.1300 and 797.1400. The 48-hour daphnia test result shall be reported and shall not be less than 1000 mg/L. The 96-hour test result for rainbow trout, bluegill or fathead minnow shall be reported and shall not be less than 1000 mg/L. This information is to be provided with bid submittal.
- 1.9. The vendor shall provide a report of The Fluid's biochemical oxygen demand (BOD) and theoretical total oxygen demand (TOD). The report shall be by an independent party and state compliance with EPA test methods. This information is to be provided with bid submittal.
- 1.10. The vendor shall provide an independent third party test report of The Fluid's compatibility with asphalt concrete pavement materials, including bitumen softening and asphalt concrete adhesion strength before-and after immersion. This information is to be provided with bid submittal.
- 1.11. The Fluid shall be stable and shall not separate in storage.

2. SPECIFICATIONS FOR SOLID RUNWAY DEICER:

- 2.1. Compound, Solid Deicing/Anti-icing for Runways and Taxiways.
- 2.2. 97% Anhydrous Sodium Acetate Base - in addition to requirements set by AMS 1431. A full copy of the latest revision of the AMS 1431 specification can be found at SAE www.sae.org
- 2.3. Specifications cover an anhydrous sodium acetate based deicing/anti-icing solid meeting FAA-approved specifications SAE AMS 1431 ("The Product").
- 2.4. The Product shall be anhydrous sodium acetate and contain no more than 1% corrosion inhibitors.

- 2.5. The vendor shall provide a copy of independent laboratory certification of the Product to AMS 1431 certification and shall be dated within thirty-six (36) months of the bid submittal.
- 2.6. The Product shall be spherical in shape.
- 2.7. The vendor shall provide comprehensive technical information covering at minimum: The Product's environmental impact, SDS, handles, storage and application guidelines with the bid submittal.
- 2.8. The vendor shall provide chemical toxicity testing for the Product. The report shall be from an independent party and meet the requirements of EPA Method 40 Code of Federal Regulations (CFR) Part 797.1300 and 797.1400. The results shall be reported and show an LC50 of greater than 1000 mg/L for the species tested.
- 2.9. The vendor shall report the Product's biological oxygen demand and toxicity with the bid submittal.
- 2.10. The vendor shall provide comprehensive on-site training to government personnel. The training will include a general overview to include application rates and usage guidelines.
3. DELIVERY: The contractor shall deliver supplies as follows:
 - 3.1. PLACE OF DELIVERY:
 - 3.1.1. The contractor shall deliver the liquid deicer to the liquid storage tanks located at building 30029, Area A, Wright-Patterson AFB, Ohio, 45433.
 - 3.1.2. The contractor shall deliver the solid runway deicer to building 30247, Area A, Wright-Patterson AFB, Ohio, 45433.
 - 3.2. TIME OF DELIVERY:
 - 3.2.1. Each delivery shall be set up for the time and place agreed upon at the time of the call by authorized personnel. Delivery shall be started within forty-eight (48) hours after placement of call and completed within seven (7) days. After the call is placed the contractor shall deliver to the assigned location as instructed.

3.2.1.1. Authorized Personnel:

Greg Wheelock
Harold Honeycutt
Travis Behnken
Aime Haas
Kristian Wade

3.3. HOURS OF DELIVERY:

3.3.1. Deliveries shall be made between 0730 and 1530 hours, unless otherwise specified by the Government due to emergency requirements, Monday through Friday.

3.4. ADVERSE WEATHER CONDITIONS:

3.4.1. If adverse weather conditions preclude delivery at the agreed upon time, the contractor shall advise an authorized personnel of the earliest possible time of delivery. In no event shall delivery be completed later than seven (7) days after placement of the call.

3.5. EMERGENCY ORDERS:

3.5.1. Emergency requirements shall be completely delivered within twenty-four (24) hours of notification, including holidays and weekends.

3.6. REFUSAL OF PRODUCT:

3.6.1. Refusal of product will be made by the 88 CES authorized personnel who is designated to receive it. Grounds for refusal include, but are not limited to:

3.6.1.1. Shipments contaminated with anything other than what is listed on the shipping label.

3.6.1.2. Ripped or damaged sacks or containers.

3.6.2. The contractor shall be responsible for transportation of refused product back to the supplier.

4. RECOGNIZED HOLIDAYS:

4.1. The contractor is not required to provide product on the following days, unless order is determined to be an emergency:

- New Year's Day
- Martin Luther King Day
- Presidents' Day
- Memorial Day
- Juneteenth
- Independence Day
- Labor Day
- Columbus Day
- Veterans' day
- Thanksgiving Day
- Christmas Day

5. Ombudsman

- a. An ombudsman has been appointed to hear and facilitate the resolution of concerns from offerors, potential offerors, and others for this acquisition. When requested, the ombudsman will maintain strict confidentiality as to the source of the concern. The existence of the ombudsman does not affect the authority of the program manager, contracting officer, or source selection official. Further, the ombudsman does not participate in the evaluation of proposals, the source selection process, or the adjudication of protests or formal contract disputes. The ombudsman may refer the interested party to another official who can resolve the concern.
- b. Before consulting with an ombudsman, interested parties must first address their concerns, issues, disagreements, and/or recommendations to the contracting officer for resolution. Consulting an ombudsman does not alter or postpone the timelines for any other processes (e.g., agency level bid protests, GAO bid protests, requests for debriefings, employee-employer actions, contests of OMB Circular A-76 competition performance decisions).
- c. If resolution cannot be made by the contracting officer, the interested party may contact the ombudsman, [Insert names, addresses, telephone numbers, facsimile numbers, and e-mail addresses of Center/MAJCOM/FLDCOM/DRU/DAFRCO ombudsman/ombudsmen]. Concerns, issues, disagreements, and recommendations that cannot be resolved at the Center/MAJCOM/FLDCOM/DRU ombudsman level, may be brought by the interested party for further consideration to the Department of the Air Force ombudsman, Associate Deputy Assistant Secretary (ADAS) (Contracting), SAF/AQC, 1060 Air Force Pentagon, Washington DC 20330-1060, phone number (571) 256-2395, facsimile number (571) 256-2431.
- d. The ombudsman has no authority to render a decision that binds the agency.

- e. Do not contact the ombudsman to request copies of the solicitation, verify offer due date, or clarify technical requirements. Such inquiries shall be directed to the contracting officer.

6.ELIMINATION OF USE OF CLASS I OZONE DEPLETING SUBSTANCES (ODS)

- a. Contractors shall not:
 - 1. Provide any service or product with any specification, standard, drawing, or other document that requires the use of a Class I ODS in the test, operation, or maintenance of any system, subsystem, item, component, or process; or
 - 2. Provide any specification, standard, drawing, or other document that establishes a test, operation, or maintenance requirement that can only be met by use of a Class I ODS as part of this contract/order.
- b. For the purposes of Department of the Air Force policy, the following products that are pure (i.e., they meet the relevant product specification identified in [AFI 32-7086\[AR2\]](#)) are Class I ODSs:
 - 1. Halons: 1011, 1202, 1211, 1301, and 2402;
 - 2. Chlorofluorocarbons (CFCs): CFC-11, CFC-12, CFC-13, CFC-111, CFC-112, CFC-113, CFC-114, CFC-115, CFC-211, CFC-212, CFC-213, CFC-214, CFC-215, CFC-216, and CFC-217, and the blends R-500, R-501, R-502, and R-503; and
 - 3. Carbon Tetrachloride, Methyl Chloroform, and Methyl Bromide.